Open-Ended Quadrupedal Locomotion through Unsupervised Environment Design

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Abstract

Unsupervised Environment Design (UED) has demonstrated strong results in improving generalization through regret-driven curriculum generation, yet its application to continuous-control robotics remains limited. We present an empirical study of PLR and ACCEL applied to quadrupedal locomotion with the Unitree Go1 in MuJoCo Playground, using a sparse, minimally engineered reward signal over a diverse procedural terrain generator. We evaluate whether these methods can construct effective curricula in this setting and whether the resulting policies exhibit robustness competitive with reward-shaped baselines. Our results demonstrate that ACCEL constructs curricula that expand beyond the support of the initial training distribution, yielding superior performance on high-complexity terrain geometries relative to a Domain Randomization baseline. Furthermore, minimally-shaped UED policies match or exceed the physical perturbation robustness of agents trained under MuJoCo Playground's dense reward-shaping scheme - under low-friction surfaces, unmodeled payloads, and actuator noise. These findings support UED as a viable training paradigm for continuous-control legged locomotion, while highlighting open challenges in scaling regret-driven methods to high-dimensional robotic control.

1 Introduction

Training legged robots to locomote robustly across unstructured terrain requires a difficult process of carefully shaping the reward function. In order to enforce stability through dense reward shaping, we make sacrifices on generalization. In the efforts to reduce reward shaping, prior work has looked into manually designed curriculum to teach locomotion with the help of environment design.

In their seminal work, Heess et al. (2017) demonstrated that sophisticated locomotion behaviours can emerge from a minimal forward-velocity reward when agents are trained across procedurally generated, obstacle-rich terrains. However, while the terrains themselves are procedurally generated, the progression of difficulty typically relies on manually tuned schedules. Such curricula are fundamentally bounded by human design, creating a "task ceiling" that limits the potential for truly open-ended behavioural discovery.

Rather than relying on environmental complexity, Hwangbo et al. (2019) achieved hardware-ready gaits by shifting the complexity into the objective function through intricate reward shaping. This paradigm is further exemplified in modern benchmarks like MuJoCo Playground (Zakka et al., 2025), where policies utilize dozens of auxiliary penalty terms, such as foot clearance and pose constraints to enforce a stable gait. While effective for producing reliable motion, these shaped rewards are embodiment-specific and prone to "specification gaming," where the agent optimizes for numerical artifacts of the reward function rather than true physical robustness. By over-constraining the gait, reward shaping often prematurely converges the agent to a single mode of locomotion, sacrificing the versatility found in emergent approaches.

Our work leverages JaxUED (Coward et al., 2024) and MuJoCo Playground to investigate whether regret-based Unsupervised Environment Design (UED) algorithms, specifically Prioritized Level Replay (PLR) (Jiang et al., 2021) and ACCEL (Parker-Holder et al., 2022) can enable open-ended locomotion learning across complex terrains with minimal reward shaping, tested on the Unitree Go1 quadruped. Our key contributions are threefold. First, we demonstrate that UED-trained agents achieve superior performance on difficult terrain geometries compared to a Domain Randomization (DR) baseline. Second, we

show that these minimally-shaped policies trained with a simple forward-velocity objective match or exceed the physical perturbation robustness of agents trained under MuJoCo Playground's dense reward-shaping scheme. Third, we provide an open empirical analysis of applying UED to continuous quadruped control, a setting that has received far less attention than discrete or 2D locomotion benchmarks. Finally, we identify limitations of our approach, including constraints imposed by the fixed network architecture and the restriction to a single quadruped morphology, rather than a broader class of legged robot embodiments.

2 Related Work

Curriculum learning in legged locomotion. Heess et al. (2017) demonstrated that procedurally varied environments can act as implicit behavioural priors, though their manually thresholded difficulty tiers scale poorly to complex parameter spaces. Several subsequent works address this by adapting terrain difficulty to agent progress. Lee et al. (2020) maintain a particle filter over terrain parameter vectors, resampling towards configurations with mid-range traversability as an automatic curriculum signal. Rudin et al. (2021) similarly introduce a game-inspired curriculum that progressively moves agents to harder terrain tiles based on performance, enabling ANYmal to learn robust locomotion via massively parallel simulation. While effective, these and related approaches such as LP-ACRL (Li et al., 2026) rely on task-specific heuristics and a manually chosen threshold rather than a signal grounded in the agent's own learning dynamics, motivating the use of more principled curriculum generation methods.

Unsupervised Environment Design. Unsupervised Environment Design (UED) encompasses a family of methods that use an underspecified environment to produce fully specified environments that enable continual learning of a policy. POET (Wang et al., 2019) pioneered open-ended environment co-evolution via minimal-criteria filtering, while PAIRED (Dennis et al., 2020) formalised this as a minimax regret game using a learned adversarial teacher. Building on these foundations, regret-based algorithms replace the learned adversary with more tractable curriculum signals: PLR (Jiang et al., 2021) maintains a replay buffer of levels prioritised by a regret proxy derived from the policy's value function, while ACCEL (Parker-Holder et al., 2022) extends PLR by applying evolutionary edits to high-regret levels to continuously expand the training frontier. While UED has demonstrated strong results in discrete and 2D domains, its application to continuous-control legged locomotion remains largely unexplored. We build on JaxUED (Coward et al., 2024), a hardware-accelerated JAX implementation of these UED algorithms, and apply PLR and ACCEL to continuous quadrupedal locomotion.

Reward Shaping for legged locomotion. Standard legged locomotion heavily relies on dense, hand-crafted reward functions to induce natural gaits. Typical formulations combine velocity-tracking terms with numerous regularizers on posture, foot slip, torque usage, and action smoothness to bias learning towards stable, energy-efficient locomotion (Hwangbo et al., 2019). Recent frameworks go further: MuJoCo Playground (Zakka et al., 2025), for example, defines dozens of auxiliary terms, like feet airtime, clearance, phase, slip, pose deviation, and base-motion penalties, whose weighted sum produces highly reliable gaits across a range of morphologies. Although these methods are highly effective, the resulting objectives are embodiment-specific, require substantial manual tuning, and can be vulnerable to specification gaming, motivating alternatives that rely less on detailed shaping and more on environment design.

3 Methodology

3.1 Environment Formulation

We model the locomotion task as an *underspecified* Partially Observable Markov Decision Process (UPOMDP) (Dennis et al., 2020). A standard Markov Decision Process (MDP) (Sutton and Barto, 1998) is defined as a tuple $\mathcal{M} = (\mathcal{S}, \mathcal{A}, T, R, \gamma)$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, $T : \mathcal{S} \times \mathcal{A} \to \mathcal{S}$ is the transition function, $R : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ is the reward function, and $\gamma \in [0, 1)$ is the discount factor. In a POMDP, the agent does not have direct access to the full state $s \in \mathcal{S}$; instead, it receives an observation $o \in \mathcal{O}$ drawn from an observation function $\Omega : \mathcal{S} \to \mathcal{O}$, which in our case consists of proprioceptive signals and privileged context provided by the MuJoCo Playground environment (Zakka

et al., 2025), but may not capture the full terrain geometry.

Our setting is further *underspecified*: rather than a single fixed MDP, we define a distribution over MDPs parameterised by a level parameter vector $\Theta \in \mathcal{T}$, yielding the tuple $\mathcal{M}_\Theta = (\mathcal{S}, \mathcal{A}, T_\Theta, R, \gamma)$. Each Θ instantiates a distinct terrain topology - specifying the configuration of obstacles such as hurdles, gaps, platforms, and hills while the reward function R and agent embodiment remain fixed. The goal is to learn a policy $\pi : \mathcal{O} \to \mathcal{A}$ that performs robustly across the full distribution $p(\Theta)$. We adopt the standard UED formulation of an underspecified POMDP, which naturally motivates the use of regret-based autocurricula to optimise the training distribution $p(\Theta)$.

3.2 Procedural Track Generation

To evaluate the agent's generalization capabilities, we define a continuous procedural terrain generator. The environment consists of a 128×128 heightfield grid, partitioned into 8 sequential segments, with all obstacles arranged along a single axis of progression (x-axis). This design deliberately isolates straight-line locomotion from lateral manoeuvring, allowing us to evaluate terrain generalization independently of steering behaviour. The terrain topology is parameterized by a continuous vector $\mathbf{p} \in [0, 1]^8$, which dictates the density and severity of four distinct topological features (Fig 1), taking inspiration from Heess et al. (2017):

- **Hills:** Continuous sinusoidal perturbations regulating terrain unevenness.
- **Platforms:** Elevated discrete flat surfaces arranged as symmetric pyramid stairs.
- **Hurdles:** Narrow vertical obstacles requiring high foot clearance, capped at 0.25m to respect the physical limits of the Unitree Go1.
- **Gaps:** Trenches of variable width and depth, testing the agent's reach kinematics.

Under the ACCEL curriculum, the blueprint $\mathbf{p}$ is iteratively updated via Gaussian mutation $\mathbf{p}' = \text{clip}(\mathbf{p} + \mathcal{N}(0, 0.1 \cdot \sigma), 0, 1)$ to construct regret-maximising obstacle sequences. A safe spawn zone is strictly enforced for the initial segment. It forces a flat surface for the first segment so that the agent learns to walk on flat ground before encountering the obstacles.

3.3 Curriculum Algorithms

3.3.1 Regret and its Approximation

The objective of UED is to train a policy π that is robust across the distribution of levels $p(\Theta)$ by solving the minimax regret problem (Dennis et al., 2020). The regret of a policy π on a level Θ is defined as:

$$\text{Regret}(\pi, \Theta) = \max_{\pi'} V^{\pi'}(\Theta) - V^{\pi}(\Theta) \tag{1}$$

Since computing true regret requires knowledge of the optimal policy, which is intractable in complex continuous-control domains, we approximate it using *Positive Value Loss* (PVL) (Jiang et al., 2021), defined as the mean positive advantage over a trajectory τ of length T:

$$\text{PVL}(\tau) = \frac{1}{T} \sum_{t=1}^{T} \max\left(0, \hat{A}_t\right) \tag{2}$$

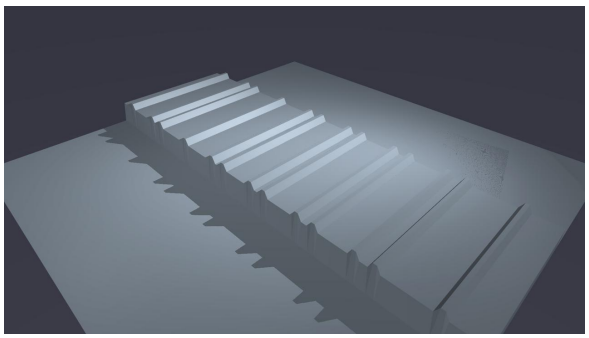
where $\hat{A}_t$ is the GAE advantage estimate at timestep t. Positive advantages indicate states where the agent performed worse than expected, signalling unresolved learning potential on that level.

3.3.2 Prioritized Level Replay (PLR)

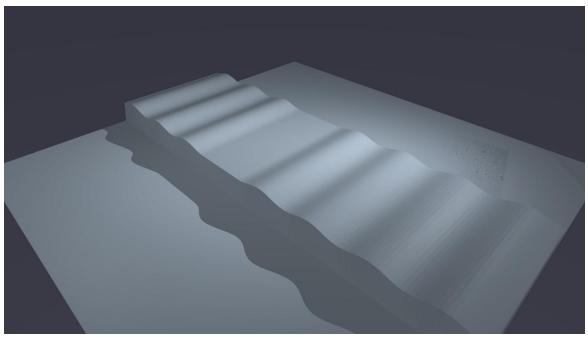
PLR (Jiang et al., 2021) maintains a fixed-capacity replay buffer of previously encountered levels, each scored by its PVL estimate. At each training iteration, a binary decision is made: with probability $1 - \rho$ a new level is sampled from the random generator and inserted into the buffer; with probability ρ a level is replayed from the buffer, sampled proportional to its PVL score weighted by a staleness coefficient to prevent stale levels from dominating.

3.3.3 ACCEL

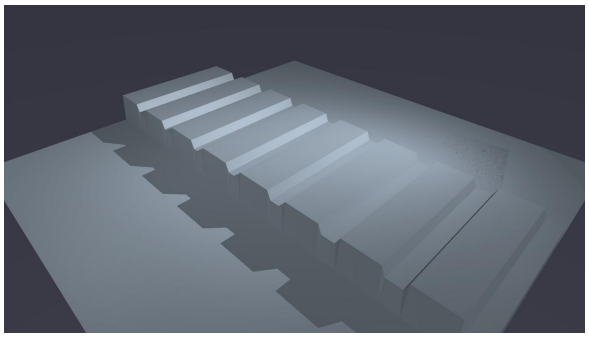
ACCEL (Parker-Holder et al., 2022) extends PLR with a third update branch: after a replay step, high-regret levels are mutated via Gaussian perturbation (Section 3.2) to produce child levels, which are evaluated and inserted into the buffer if their PVL score remains high. The training loop therefore cycles between three phases: new level generation, replay, and mutation. Critically, while ACCEL is initialized identically to PLR,



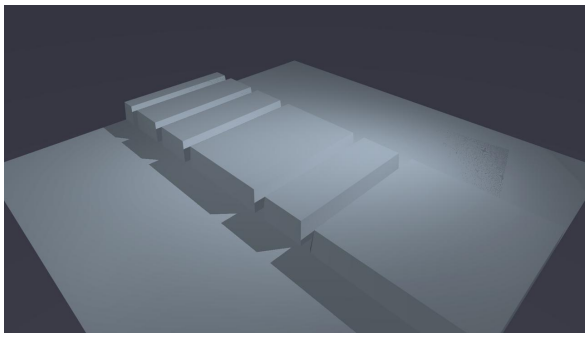
(a) Hurdles



(b) Hills



(c) Platforms



(d) Gaps

Figure 1: Illustrative tracks with obstacles

its mutation step applies unbounded perturbations to high-regret levels, allowing it to autonomously breach the initial 0.6 density ceiling and invent complex obstacle combinations that the random generator was forbidden from creating. To further stabilise early training, we additionally enforce a 10% probability of sampling a fully flat level during new level generation.

3.3.4 Domain Randomization Baseline

As a baseline we compare against Domain Randomization (DR) (Tobin et al., 2017), which samples Θ uniformly from $\mathcal{T}$ at each episode with no adaptive curriculum signal, allocating equal training budget regardless of level informativeness. DR corresponds to the degenerate case of PLR with replay probability $\rho = 0$.

3.4 Minimal Reward Formulation

A central hypothesis of this work is that Unsupervised Environment Design (UED) can substitute for dense kinematic reward shaping. Consequently, we omit standard quadrupedal shaping terms such as foot clearance trajectories, symmetric phase timing. The agent optimises a minimal reward function defined primarily by forward task progress, track geometry, and a survival prior.

At each timestep t, the reward $\mathcal{R}_t$ is defined as:

$$\mathcal{R}_t = \begin{cases} -10.0 & \text{if fallen} \\ \bar{r}_t + 10.0 & \text{if reached goal} \\ \bar{r}_t & \text{otherwise} \end{cases} \tag{3}$$

where $\bar{r}_t = r_{\text{task}} + r_{\text{reg}} + r_{\text{lazy}}$.

where the individual components are defined as:

$$r_{\text{task}} = \text{clip}(v_x,\ 0.0,\ 2.5) \tag{4}$$

$$r_{\text{reg}} = -2.0 \cdot \mathcal{P}_{\text{center}} - 0.05 v_y^2 - 1.0(q_x^2 + q_z^2) - 0.05\omega_z^2 \tag{5}$$

$$r_{\text{lazy}} = \begin{cases} -1.0 & \text{if } v_x < 0.2 \\ 0.0 & \text{otherwise} \end{cases} \tag{6}$$

It is important to note that the regularisation terms in r_{reg} are *not* behavioural shaping terms in the conventional sense. Rather, they are geometric constraints required to keep the robot oriented safely on the 4.0m-wide physical track: since the agent must traverse a narrow corridor aligned with the x-axis (Section 3.2), lateral drift (v_y), off-centre displacement (y) - defined as $\mathcal{P}_{\text{center}} = y^2 + 10.0 \max(0, |y| - 0.8)^2$ - and yaw rotation (ω_z) must be penalised to prevent the robot from

walking off the track entirely. These terms encode the geometry of the task, not a prior over gait style. The only term that could be considered a behavioural prior is the nose error e^2_{nose}, which discourages the robot from pitching forward, and the action penalty $\|\mathbf{a}\|^2$, which is kept to avoid explosive torque spikes and protect the robot.

Calibrating these geometric constraints required balancing track adherence against navigational freedom. Early iterations demonstrated that overpenalising lateral kinematics induced policy paralysis; the agent would fail to execute necessary steering corrections and ultimately drift off the track edge. To resolve this optimisation conflict, we substantially reduced the penalty coefficients for lateral drift (v_y) and yaw rate (ω_z). By shifting the primary containment pressure to the dual-tier $\mathcal{P}_{\text{center}}$ boundary, we mathematically permit the agent to momentarily absorb minor penalty costs to execute fluid trajectory adjustments without triggering catastrophic early-episode termination.

Finally, the posture penalty targets the torso's roll (q_x) and yaw (q_z) quaternions, forcing the agent to maintain a level spine relative to the track. Crucially, we deliberately exclude pitch (q_y) from this penalty to allow the agent to rear up when climbing hurdles. By reducing explicit torque penalties ($\|\mathbf{a}\|^2$) while enforcing this strict roll constraint, we provided the agent with the mechanical freedom to execute jumps.

3.5 Policy Architecture

Unlike the hierarchical, pre-trained spinal and cortical sub-networks required to achieve emergent locomotion in complex environments (Heess et al., 2016, 2017), we employ a single, non-hierarchical recurrent Actor–Critic architecture trained end-to-end with PPO. The agent observes a unified *privileged state* vector, which concatenates proprioceptive and exteroceptive signals from MuJoCo Playground into a 60-dimensional input that is fed directly to the policy network.

The privileged state is first processed by a shared feedforward embedding layer: a single fully connected layer with 256 units and ReLU activation. This embedding is passed into a recurrent memory core implemented as an Optimized LSTM cell with 256 hidden units, wrapped in a reset mechanism that zeroes the hidden state on episode termination. The LSTM output then branches into decoupled Actor and Critic heads. Each head consists of two fully connected layers with 256 and 128 units (ReLU), followed by a task-specific output layer: the Actor outputs a 12-dimensional mean action vector and a learned log-standard-deviation parameter to define a multivariate normal distribution over joint torques, while the Critic outputs a scalar state-value estimate to support GAE-based value learning.

4 Experimental Setup

To empirically validate our hypotheses, we design a rigorous experimental framework using the MuJoCo Playground physics engine, accelerated via JAX. The experiments are partitioned into two distinct evaluation suites: Suite I isolates geometric generalization, while Suite II isolates dynamic physical robustness.

4.1 Implementation Details and Baselines

The Unitree Go1 agent is trained end-to-end using Proximal Policy Optimization (PPO) coupled with Generalized Advantage Estimation (GAE). Each training episode consists of a maximum of 1,500 simulation steps, providing the agent sufficient time to traverse the track. During UED training (PLR and ACCEL), the environment sampler evaluates regret primarily via Positive Value Loss (PVL).

We evaluate the proposed autocurriculum methods against three representative control baselines:

- **Baseline 1 (Minimal reward, No obstacles):** Trained exclusively on flat terrain utilizing our minimal reward formulation.
- **Baseline 2 (Dense reward, No obstacles):** Trained on flat terrain but utilizing exhaustive, hand-crafted kinematic shaping terms (e.g., strict joint pose tracking, foot clearance minimums, and aggressive energy penalties) adapted from the standard DeepMind control suite.
- **Baseline 3 (Minimal reward, Domain Randomization):** Trained using minimal rewards on procedurally generated tracks where obstacle parameters are sampled uniformly at random, lacking a regret-driven curriculum structure.

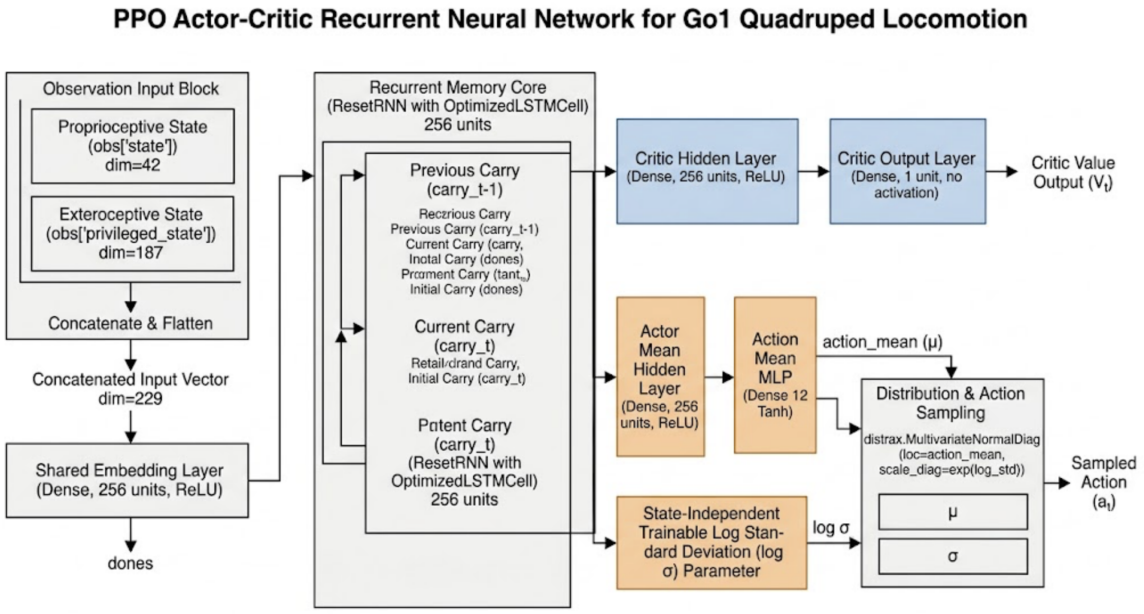


Figure 2: The recurrent continuous Actor–Critic architecture. The privileged state (proprioceptive and exteroceptive signals) is embedded by a shared MLP, processed by an Optimized LSTM core, and then passed to decoupled Actor and Critic heads.

4.2 Evaluation Metrics

Sparse reward reinforcement learning is a challenge where agents receive infrequent, delayed feedback, typically only upon task completion, making exploration inefficient and learning slow. Because binary success metrics frequently fail to capture partial progress in highly complex environments, we evaluate the policies using a normalized distance metric.

Normalized Distance (%): A continuous progress metric ($0.0 \rightarrow 1.0$) measuring the maximum forward global distance covered before episode termination (due to time limits or terminal falls), relative to the total 34.0 m track length. We start our track from $X = -18.0$ m and measure the finish line at $X = 16.0$ m.

4.3 Suite I: Policy Evaluation on Test set

To assess the policies' ability to navigate complex spatial topologies, the agents are evaluated on a fixed suite of 300 procedurally generated tracks sampled from the full parameter space to ensure a comprehensive evaluation across the difficulty spectrum. The suite is stratified into three tiers: Tier 1 (100 Easy tracks with minimal variance), Tier 2 (100 Medium tracks featuring isolated hurdles and gaps), and Tier 3 (100 Hard tracks presenting dense and more difficult obstacle clusters). This suite evaluates the policy on the test set, comparing the performance of ACCEL, PLR and Domain Randomization.

Track difficulty is operationalised through a set of procedural generation parameters governing obstacle density, height, gap width, and platform spacing.

- **Easy tracks:** Sampled from a restricted region of this parameter space, with low obstacle density, bounded height variation and no gap obstacles.
- **Medium tracks:** Introduce isolated hurdles and gaps drawn from intermediate parameter ranges, requiring deliberate stepping adjustments but presenting no complex obstacle sequences.
- **Hard tracks:** Sampled from the full upper range of the parameter space, featuring dense obstacle clusters, larger height discontinuities, and wider gaps that demand coordinated multi-step recovery behaviours.

Specific parameter ranges were selected to ensure each tier presented qualitatively distinct challenges, as verified by inspection of generated tracks.

4.4 Suite II: Dynamic Robustness

To test whether regret-driven autocurricula instill better physical robustness than dense reward shaping, we evaluate the agents under severe dynamic perturbations. To isolate recovery behaviours from obstacle navigation (avoiding collision-based failures), these tests are conducted on perfectly flat terrain across four conditions:

- **Control:** Standard MuJoCo physics parameters on a high-friction surface.
- **Ice Test:** Substrate sliding friction drastically reduced from $\mu \approx 1.0$ to $\mu = 0.1$, testing the agent's ability to dynamically widen its stance to prevent lateral slip.
- **Payload Test:** An unobserved $10.0\,\text{kg}$ mass is rigidly attached to the robot's torso, testing torque adaptation and center-of-mass stabilization.
- **Actuator Noise:** 25% random Gaussian noise is continuously injected into the network's action outputs before application to the physics engine, evaluating the policy's real-time error-correction capabilities under severe actuation degradation.

5 Results and Discussion

5.1 Curriculum Trajectory and Autonomous Difficulty Scaling during Training

Figure 3 presents the mean returns and learning curves (defined as Normalized Distance % under section 4.2) across the three difficulty tiers, evaluated periodically throughout training on representative Easy, Medium, and Hard tracks. On Easy and Medium tracks, all three training strategies achieve comparable performance, suggesting that any reasonable training distribution is sufficient to tackle isolated, low-density obstacles. However, on Hard tracks - featuring dense obstacle clusters and complex terrains - ACCEL starts outperforming both PLR and DR.

This result directly validates our central hypothesis. DR samples uniformly across the task distribution space, meaning hard configurations appear only rarely in the training distribution and receive no preferential treatment. PLR prioritizes high-regret levels but relies entirely on the random generator to produce them, limiting its exposure to more complex topologies. ACCEL, by contrast, autonomously discovers and amplifies hard configurations through evolutionary mutation of high-regret levels, organically constructing the dense obstacle sequences required to train competent performance on Tier 3 tracks.

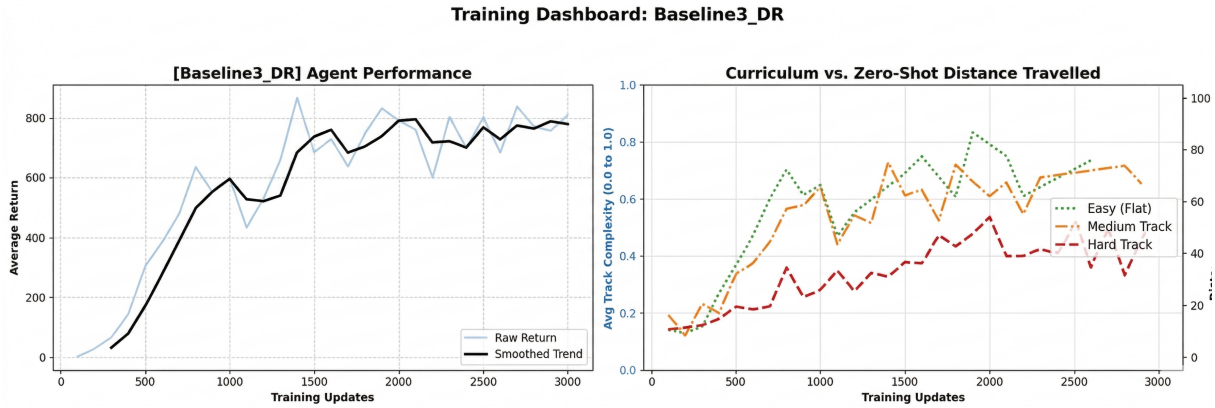
Figure 3 further reveals an initial plateau in PLR's performance on Medium and Hard tracks, persisting for approximately 1,200 training updates before meaningful improvement emerges. Unlike DR, which benefits from uniform sampling across the distribution from the outset, PLR commits to a replay buffer whose prioritisation is unreliable early in training due to noisy regret estimates. As a result, PLR over-replays an arbitrary subset of levels during this initial phase, delaying productive curriculum progression until the value function stabilises. ACCEL exhibits no such plateau, as its mutation operator continuously generates frontier-level configurations regardless of buffer maturity.

This observation aligns with other recent works (Rutherford et al., 2024) that have shown that standard regret approximations (e.g., PVL / MaxMC) can fail to identify genuinely learnable levels, causing UED methods such as PLR and ACCEL to over-sample either already-solved or effectively unsolvable maps and, in some domains, to match or underperform naive Domain Randomization.

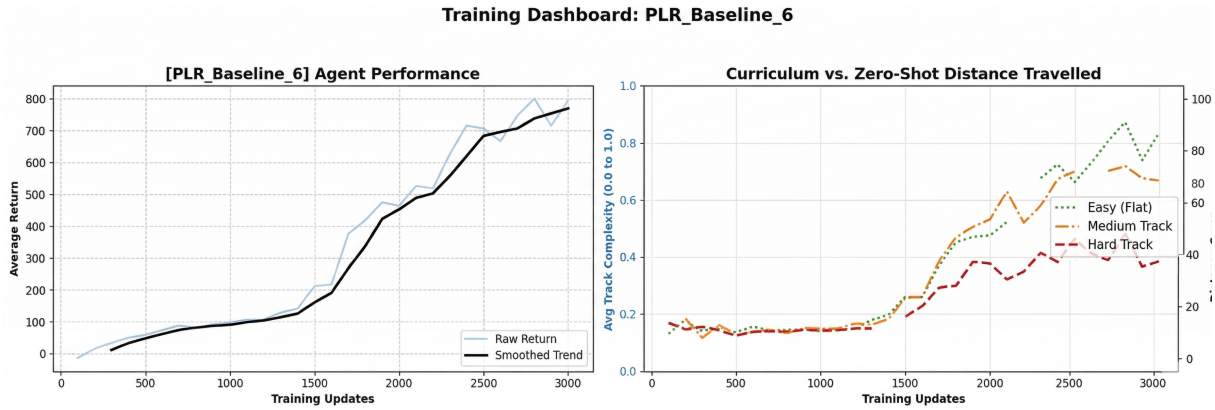
5.2 Suite I: Policy Evaluation on Test set

Table 1 details low-variance performance estimates of trained policies measured by Normalized Distance averaged over 300 Easy/Medium/Hard held-out test tracks.

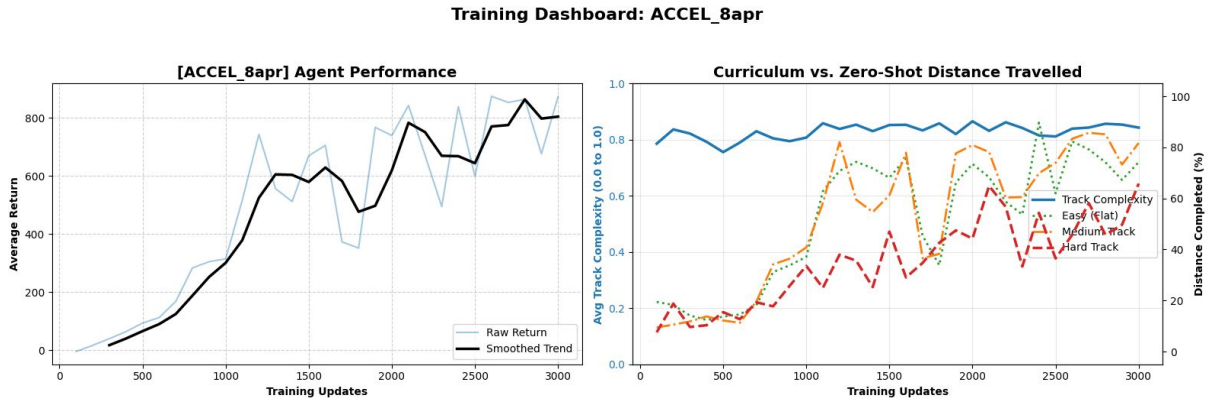
Notably, the results reveal opposing performance profiles of PLR and ACCEL. PLR achieves near-perfect performance on Easy tracks (95.8%) but collapses on Hard tracks (24.1%), consistent with its distributional ceiling: PLR can only curate levels drawn from the random generator, and the dense obstacle clusters present in the curated Hard set (see 4.3) lie largely outside its training support. ACCEL, conversely, excels on Hard tracks (59.1%) but underperforms on Easy (58.6%) tracks. As the ACCEL curriculum matures, levels with low regret (already-solved easy configurations) are continuously displaced from the replay buffer in favour of higher-regret frontier levels, reducing the agent's exposure to simple tracks late in training. DR achieves the most balanced profile, with no tier excelling but also no



(a) DR learning curve



(b) PLR learning curve



(c) ACCEL learning curve

Figure 3: Training Dashboard comparison for UED policies using (a) DR and (b) PLR. Both dashboards display (Left) Mean episode return over training updates, and (Right) Continuous Normalized Distance tracking early-stage task progression.

catastrophic failure, reflecting the consistent coverage guaranteed by uniform sampling.

Policy	Overall%	Tier		
		Easy	Med	Hard
Baseline 3 (DR)	64.2%	71.2%	75.4%	48.9%
PLR	61.0%	**95.8%**	76.9%	24.1%
ACCEL (Ours)	**65.1%**	58.6%	75.0%	**59.1%**

Table 1: Suite I Results: Solve rate (Normalised Distance %) across the 300-track evaluation Suite I.

5.3 Suite II: Dynamic Robustness

Figure 4 details low-variance robustness scores of trained policies measured by Normalized Distance averaged over test tracks with varying physics (100 each).

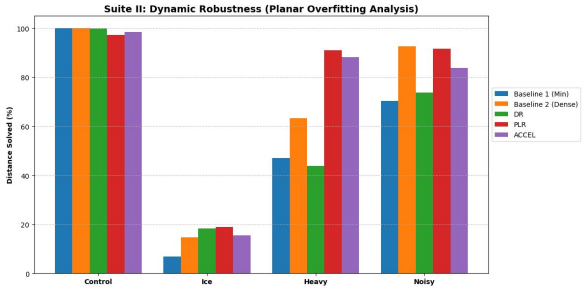


Figure 4: Suite II results: Solve rate (Normalised Distance %) across the 400-track evaluation Suite II.

The results of Suite II (Figure 4) strongly validate our central hypothesis regarding the brittleness of explicit reward shaping. Under the standard Control physics, Baseline 2 (Dense Shaping) achieved near-perfect traversal, converging on an aesthetically natural walking gait. However, when introduced to physical perturbations, the policy performance degrades.

Because the heavily shaped agent was optimized strictly to satisfy kinematic constraints on a mathematically perfect, flat topography, it lacked the dynamic spatial awareness required to actuate corrective torques under a 10.0 kg payload, or widen its stance on ice ($\mu = 0.1$).

By contrast, the proposed UED policies, operating primarily without kinematic shaping, maintained high traversal distances across all perturbation tests except reduced friction. By constantly failing and recovering from navigating tougher terrains during its UED training, the agent rewired its policy to generate more complex gait. This can be seen by the results: UED performs similar to the dense reward policy across various perturbations. This demonstrates that environmental pressure via UED acts as a promising approach to enable generalisation compared to exhaustive, state-based reward engineering.

5.4 Qualitative Gait Analysis

Visual inspection of showcase rollouts reveals distinct behavioral morphologies across methods. The minimal-reward, obstacle-free baseline (Baseline 1) converges to an erratic high-frequency gait sufficient for flat terrain but incapable of handling complex topologies or dynamic perturbations.

The dense-reward baseline (Baseline 2) produces a visually perfect symmetric trot. This doesn't always generalise across unseen environments.

Domain randomization with minimal rewards (Baseline 3) yields a more tactile gait capable of moderate terrain complexity, though it remains unstable on hard tracks and fails robustness tests.

Both PLR and ACCEL develop specialized leaping-bound gaits. While PLR's gait lacks universal terrain adaptability, ACCEL demonstrates stride length adjustment and centre-of-mass lowering when approaching hurdles, alongside stance widening under payload and low-friction perturbations. These targeted adaptations suggest UED enables more flexible recovery behaviours than rigid shaping, despite similar overall gait morphology.

6 Limitations and Future Work

Reward Formulation. Although our reward function is deliberately minimal, it still relies on a positive x-velocity bonus to incentivise forward progress rather than explicit velocity tracking. A target-velocity formulation would provide a more principled locomotion objective and allow finer control over gait speed. Additionally, the current track geometry exposes a boundary failure mode: agents occasionally drift laterally and fall off the track edge, suggesting that the geometric containment penalties in r_{reg} may require further calibration.

Single Morphology. All experiments are conducted exclusively on the Unitree Go1 quadruped. It remains an open question whether regret-driven autocurricula generalise to morphologies with fundamentally different stability margins, such as bipedal humanoids, where the reduced base of



Figure 5: Qualitative Gait Analysis of the ACCEL policy. (Left) The agent dynamically lowers its center of mass to prepare for the jump. (Center) Synchronized hind-leg propulsion to clear the 0.25m hurdle. (Right) Recovery during landing to absorb the impact and maintain forward momentum without heavily engineered kinematic constraints.

support makes robust recovery behaviours strictly harder to acquire. Extending this framework to diverse embodiments is a natural direction for future work.

Symmetric Actor-Critic Architecture. Our policy uses a shared privileged state for both the Actor and Critic, departing from the asymmetric actor-critic design commonly adopted in sim-to-real transfer (Pinto et al., 2017), where the Critic has access to privileged simulation state during training but the Actor is restricted to proprioceptive observations available on real hardware. Adopting an asymmetric architecture would better align our training setup with deployment constraints and is likely to improve zero-shot sim-to-real transfer performance.

Only Regret-Based UED. This work evaluates only regret-based UED algorithms (PLR and ACCEL). Alternative autocurriculum signals, such as learning progress (Portelas et al., 2020) or competence-based measures (Dennis et al., 2020), may offer complementary advantages, particularly in settings where value-function-based regret proxies are unstable or slow to converge. A systematic benchmark comparing regret-based and learning-progress-based UED methods in continuous control remains an open and worthwhile direction.

Finally, this work is currently bounded within the MuJoCo simulation framework. **While the perturbations in Suite II suggest high dynamic robustness**, the ultimate test of an unshaped, UED-trained policy is zero-shot Sim-to-Real transfer onto physical hardware. Future work will investigate whether the emergent stability learned via UED is sufficient to bridge the reality gap without the need for domain randomization of actuator dynamics or system latency.

7 Conclusion

This work revisits the role of reward design in legged locomotion by examining whether complexity can be shifted from the objective function to the training environment. Through a systematic evaluation of regret-driven Unsupervised Environment Design, we show that minimally shaped objectives, when paired with adaptive curricula, can produce robust locomotion behaviours in continuous quadrupedal control.

Our results support three main observations. First, curriculum generation driven by the agent's learning signal, particularly through ACCEL's mutation of high-regret environments enables the training distribution to progressively expand in difficulty, improving performance on challenging terrain configurations. Second, while dense reward shaping reliably induces stable and visually regular gaits, it can also introduce implicit constraints that reduce robustness under distribution shift, which can be mitigated by training a minimally shaped policy through UED methods. Third, exposure to diverse and failure-inducing environments appears to encourage the emergence of adaptive recovery strategies, suggesting that environment design can act as a meaningful inductive bias alongside, or in place of, handcrafted reward terms.

Overall, our findings suggest that regret-based environment design is a promising direction for reducing dependence on reward shaping in continuous control, while raising important questions about scalability, stability, and generalization that warrant further investigation.

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A Appendix

A.1 Hyperparameters

The policies were trained utilizing the hyperparameter configuration detailed in Table 2, executed via MuJoCo Playground's JAX-accelerated PPO implementation.

Parameter	Value
PPO Parameters	
Total Updates	3000
Num Train Envs	256
Num Steps per Update	128
Minibatches	8
Epochs per Update	4
Learning Rate	3×10^{-4}
Max Grad Norm	0.5
Clip ϵ	0.2
Gamma (γ)	0.99
GAE Lambda (λ)	0.95
Entropy Coefficient	0.005
Critic Coefficient	0.5
UED Parameters (ACCEL/PLR)	
Buffer Capacity	5000
Replay Probability	0.5
Score Function	Positive Value Loss
Prioritization	Rank-based
Top-K (k)	4
Temperature	0.1
Staleness Coefficient	0.3
Minimum Fill Ratio	0.5
Number of Mutations	1
Exploratory Grad Updates	True
Evaluation Parameters	
Evaluation Frequency (Updates)	100
Rollouts per Eval Level	10

Table 2: Hyperparameters for the continuous Actor-Critic PPO network and the UED Level Sampler.

A.2 Baseline 2: Dense Reward Shaping Terms

To provide a rigorous comparison against traditional control methods, Baseline 2 (Dense Shaping) utilized the comprehensive quadrupedal reward formulation adapted from the DeepMind control suite. In addition to the forward velocity and alive bonuses, this baseline optimized against a dense array of kinematic penalties, including:

- **Base Posture Constraints:** Penalties for roll, pitch, and Z-height deviations from a nominal standing pose.
- **Energy and Torque Limits:** L^2 penalties on joint torques and action rate-of-change to enforce smooth actuation.
- **Foot Clearance:** Tracking desired foot trajectories to enforce a symmetrical trotting gait and minimize ground-scuffing.